

Demonstration of Lenticular Objects: 3D Printed Objects with Lenticular Lens Surfaces That Can Change their Appearance Depending on the Viewpoint

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Figure 1: 3D printed objects with lenticular lens surfaces enable viewers to see different appearances from different viewpoints (a: looking down; b: looking horizontally; c: looking up). Our user interface supports designers in setting up different viewpoints and assigning the corresponding textures. On export, it automatically generates the files for fabrication. Designers can then 3D print the object geometry, lenses, and underlying color patterns in a single pass with a multi-material 3D printer.

ABSTRACT

We present Lenticular Objects, which are 3D objects that appear differently from different viewpoints. We accomplish this by 3D printing lenticular lenses across the curved surface of objects and computing underlying surface color patterns, which enables to generate different appearances to the user at each viewpoint.

In addition, we present the Lenticular Objects 3D editor, which takes as input the 3D model and multiple surface textures, i.e. images, that are visible at multiple viewpoints. Our tool calculates the lens placements distribution on the surface of the objects and underlying color pattern. On export, the user can use ray tracing to live preview the resulting appearance from each angle. The 3D model, color pattern, and lenses are then 3D printed in one pass on a multi-material 3D printer to create the final 3D object. We demonstrate our system in practice with a range of use cases that benefits from appearing differently under different viewpoints.

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UIST '22 Adjunct, October 29-November 2, 2022, Bend, OR, USA

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ACM ISBN 978-1-4503-9321-8/22/10.

<https://doi.org/10.1145/3526114.3558628>

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

multi-material 3D printing; optics; lenticular lenses; design tools.

ACM Reference Format:

Jiani Zeng, Honghao Deng, Yunyi Zhu, Michael Wessely, Axel Kilian, and Stefanie Mueller. 2022. Demonstration of Lenticular Objects: 3D Printed Objects with Lenticular Lens Surfaces That Can Change their Appearance Depending on the Viewpoint. In *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*, October 29-November 2, 2022, Bend, OR, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3526114.3558628>

1 INTRODUCTION

Lenticular printing refers to the method of using lenticular lenses to show different images from different viewpoints [5]. It is commonly applied in advertisement and arts to achieve special purpose effects, such as showing various images on one commercial motion card [3, 7] or creating artistic effects with the illusion of depth [2, 4].

Traditionally, lenticular prints are limited to 2D. They are fabricated by placing flat lenticular sheets onto 2D color patterns composed of multiple images. Depending on the viewpoint, light rays enter the lens at different angles and reflect off different parts of the underlying color pattern. Therefore, under different viewpoints,

a different part of the color pattern is visible to the viewer’s eye and thus the viewer sees only one image from the color pattern.

One reason why lenticular prints do not yet exist in the form of 3D objects is that up until recently, no fabrication process existed that was able to manufacture the lenticular lenses and high resolution color patterns on doubly curved surfaces. However, over the last years, multi-material 3D printers have been developed that can print with a multitude of materials and in high-resolution color [1]. For example, the 3D printer Stratasys J55 [6] can print with optical clear materials, which can be used for the lenses, as well as CMYK materials, which can be used to print the color pattern.

In this demonstration, we present *Lenticular Objects*, creating curved 3D objects with lenticular surfaces [8] that look different from multiple viewpoints. We demonstrate our 3D editor tool that takes as input the 3D model, the desired viewpoints, and the corresponding color textures and then computes the lens placement and color pattern across the object’s surface (Figure 1). Finally, we demonstrate the usefulness of lenticular objects with four application examples in product design and HCI.

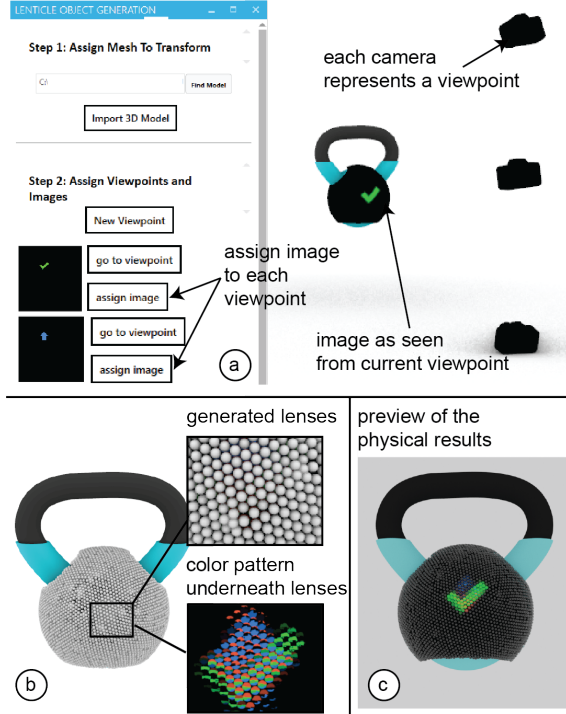


Figure 2: (a) Our 3D editor allows to create viewpoints, assign images to each viewpoint and verify assignments through the viewport. (b) The system generates the lens geometry and the underlying color patterns. (c) Before fabrication, designers can simulate the physical result using ray tracing.

2 END TO END SYSTEM

Designers use our 3D editor¹ to specify the object geometry, available viewpoints and the image for each viewpoint (Figure 2a). Our

¹code available at <https://github.com/yunyi-zhu/lenticular-object>

system then generates the corresponding 3D model of the lenses and underlying color patterns that achieves the specified inputs (Figure 2b). Before fabrication, designers can use ray-tracing to (Figure 2c) to simulate the result, which provides a more accurate preview of the object. The files are then sent to a 3D printer, which fabricates the Lenticular Object in one pass.

3 APPLICATIONS

Being able to design objects that look different from different viewpoints enables a wide range of novel applications in product design and HCI. We will present four different objects that we have generated with the Lenticular Object system, including (1) a kettlebell that directs the user to a specific body position (Figure 1), (2) a lampshade that shows personalized greetings to the viewer depending on their location (Figure 3a), (3) a shoe that shows a supportive message that is only available to the wearer (Figure 3b), and (4) an earpod case that with a dynamic product design effect (Figure 3c).

The application scenarios that we present demonstrate that our system can produce objects with: (1) geometric complexities (single and doubly curved surfaces), (2) image complexities (patterns, symbols and text), and (3) number of distinct images (ranging from 2 to 5).

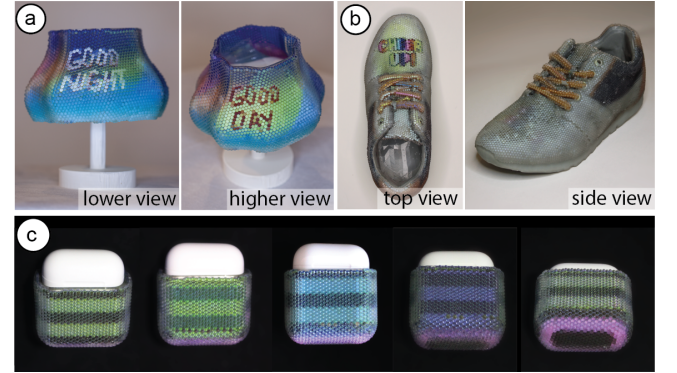


Figure 3: Application Examples. (a) A lampshade that shows different messages if the users stands or lies down, (b) a shoe with a personal message only visible to the wearer, and (c) an earpod case that shows viewpoint-dependent images.

4 CONCLUSION

In this demonstration, we presented a method to extend lenticular displays to 3D object geometries, enabling 3D objects to look different from different viewing angles. We showed how our design tool supports designers in creating 3D lenticular lens displays by assigning color textures to multiple viewpoints. We demonstrated four example applications that highlight different geometric and image complexities as well as different numbers of viewpoints.

ACKNOWLEDGMENTS

We thank Neil Gershenfeld and Tom Lutz for their help on 3D printing our models on a Stratasys J55. This paper is based upon work supported by the National Science Foundation under Grant No. 1716413.

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